

Problem 3 (Bulgarian solitaire), Cell 5: two above a triangular number

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Section 1 restates these definitions in an equivalent form, proves that $d_{\mathcal{B}}$ is well defined and that the rank exists and is unique (Lemma 1.2, where we also write $n = T_{k-1} + r$, $1 \leq r \leq k$), and fixes further notation.

Official statement of the cell.

The next family: $n = T_{k-1} + 2$, that is $n = 3, 5, 8, 12, 17, 23, \dots$ for $k = 2, 3, 4, 5, 6, 7, \dots$. Determine $D_{\mathcal{B}}(T_{k-1} + 2)$ for every k , with proof of both bounds. State exactly for which k your formula holds, and give the remaining values separately. Your upper bound must be a single argument valid for all k in that range, not a separate treatment of each k , and you must give the extremal partitions explicitly as a function of k . Say also where the straightforward extension of the C3 argument stops: give the bound it does yield, show it is strictly weaker than the truth, and identify precisely what your proof supplies in its place.

Status. Partial: value and both bounds fully proved for the stated range; the complete set of extremal partitions is given only by an exact dynamical criterion (closed form open). The formula for $k \geq 7$ is proved by one argument covering all $k \geq 7$. The values for $2 \leq k \leq 6$ come from an exhaustive computation over all partitions of $n = 3, 5, 8, 12, 17$ (code in Appendix A). Extremal partitions: an explicit extremal family for every $k \geq 7$, the complete explicit lists for $k \leq 6$, and an exact criterion for $k \geq 7$. A closed-form list of *all* extremal partitions for $k \geq 7$ is not given (there are 66, 253, 1341, $\dots$ of them), in line with Cells 2–4. If the cell requires such a list, this sub-part is partial. **New relative to Griggs–Ho:** for $k \geq 7$ this is the case $r = 2$ of their Conjecture 4.7 [4], which was open. The same caveat on novelty as in Cell 4 applies.

Answer and where it is proved.

- **Formula.** $D_{\mathcal{B}}(T_{k-1} + 2) = (k-1)(k-4)$ for every $k \geq 7$, and for no $k \leq 6$ (Theorem 9.1).

- **Remaining values.** $D_{\mathcal{B}}(T_{k-1} + 2) = 2, 3, 5, 8, 12$ for $k = 2, 3, 4, 5, 6$ (i.e. $n = 3, 5, 8, 12, 17$). In one line, $D_{\mathcal{B}}(T_{k-1} + 2) = \max\{(k-1)(k-4), T_{k-2} + 2\}$ for all $k \geq 2$.
- **Upper bound** (Theorem 7.1): a classification of how orbits enter the cycle (Lemma 5.2), the Griggs–Ho ramp bound, and a new *pile-chain lemma* (Lemma 6.1).
- **Extremal partitions.** For $k \geq 5$, $\mu_k = (k-2, k-2, k-3, k-4, \dots, 4, 3, 3, 2, 1)$ ($(\mu_k)_1 = k-2$, $(\mu_k)_i = k-i$ for $2 \leq i \leq k-3$, $(\mu_k)_{k-2} = 3$, $(\mu_k)_{k-1} = 2$, $(\mu_k)_k = 1$; e.g. $(3, 3, 3, 2, 1)$, $(4, 4, 3, 3, 2, 1)$ for $k = 5, 6$) has $d_{\mathcal{B}} = (k-1)(k-4)$, hence is extremal for $k \geq 7$ (Theorem 8.2; the witness of [4, Thm. 4.5, case (1)] with $r = 2$). For $k \geq 7$, λ is extremal iff $\mathcal{B}^{(k-1)(k-4)-1}(\lambda)$ has at most $k-2$ parts (Theorem 10.1). For $k \leq 6$ the complete lists are in Theorem 9.1.
- **The C3 argument** yields only $k^2 - 2k - 1$, which exceeds the truth by $3k - 5$. What replaces it is explained in Section 11.

Sections 1–4 are shared with Cells 2–4, and are repeated so that this document is self-contained.

1 Setting and notation

Throughout, $T_m = m(m+1)/2$ for integers $m \geq 0$, and $[P] \in \{0, 1\}$ denotes the truth value of a statement P . A *partition* λ of $n \geq 1$ (written $\lambda \vdash n$) is a finite non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell$ with sum n ; $\ell = \ell(\lambda)$ is its number of parts, and we put $\lambda_i = 0$ for $i > \ell(\lambda)$. Partitions are identified with multisets of positive integers.

Definition 1.1 (Bulgarian solitaire). For $\lambda \vdash n$ let $\mathcal{B}(\lambda)$ be the partition whose parts are $\ell(\lambda)$ together with the numbers $\lambda_i - 1$ for those $i \leq \ell(\lambda)$ with $\lambda_i \geq 2$. Since $\ell(\lambda) + \sum_i (\lambda_i - 1) = n$, we have $\mathcal{B}(\lambda) \vdash n$. Put $\mathcal{B}^0 = \text{id}$ and $\mathcal{B}^{i+1} = \mathcal{B} \circ \mathcal{B}^i$. A partition μ is *cyclic* if $\mathcal{B}^P(\mu) = \mu$ for some $P \geq 1$. For $\lambda \vdash n$, $d_{\mathcal{B}}(\lambda)$ is the least $i \geq 0$ such that $\mathcal{B}^i(\lambda)$ is cyclic, and $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \vdash n\}$. A partition $\lambda \vdash n$ with $d_{\mathcal{B}}(\lambda) = D_{\mathcal{B}}(n)$ is called *extremal*.

$d_{\mathcal{B}}(\lambda)$ is well defined: n has finitely many partitions, so $\mathcal{B}^a(\lambda) = \mathcal{B}^{a+P}(\lambda)$ for some $a \geq 0$, $P \geq 1$, and then $\mathcal{B}^a(\lambda)$ is cyclic. If μ is cyclic with $\mathcal{B}^P(\mu) = \mu$, then $\mathcal{B}^P(\mathcal{B}(\mu)) = \mathcal{B}(\mu)$, so $\mathcal{B}(\mu)$ is cyclic. Hence $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$ and for no $i < d_{\mathcal{B}}(\lambda)$.

Lemma 1.2 (rank). *Every $n \geq 1$ can be written in exactly one way as $n = T_{k-1} + r$ with integers $k \geq 1$ and $1 \leq r \leq k$. We call k the rank of n ; n is triangular iff $r = k$ (then $n = T_k$).*

Proof. $T_0 = 0 < T_1 < T_2 < \dots$ is strictly increasing and unbounded, so there is exactly one $k \geq 1$ with $T_{k-1} < n \leq T_k$; then $r = n - T_{k-1}$ satisfies $1 \leq r \leq T_k - T_{k-1} = k$. Conversely $n = T_{k-1} + r$ with $1 \leq r \leq k$ forces $T_{k-1} < n \leq T_k$, i.e. this k . If $n = T_m$ ($m \geq 1$), then $T_{m-1} < n \leq T_m$, so $k = m$ and $r = T_m - T_{m-1} = m = k$; conversely $r = k$ gives $n = T_k$. $\square$

2 Diagrams, diagonals, rotation, and Brandt’s theorem

This section follows Griggs–Ho [4, §2] (the idea goes back to Etienne [3]); we make their “left shifting decreases the diagonal index” precise with an explicit potential Φ .

Let $\mathbb{P} = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$; the *cell* $(i, j) \in \mathbb{P}$ has *row* i and *column* j . The *diagram* of λ is

$$Y(\lambda) = \{(i, j) \in \mathbb{P} : 1 \leq j \leq \ell(\lambda), 1 \leq i \leq \lambda_j\},$$

so column j has λ_j cells (columns are parts, as in [4]). $Y(\lambda)$ determines λ , since $\lambda_j = \#\{(i, j) \in Y(\lambda)\}$. Row 1 of $Y(\lambda)$ is $\{(1, j) : j \leq \ell(\lambda)\}$ and column 1 has λ_1 cells; in particular

$$(1, w+1) \in Y(\lambda) \implies (1, w) \in Y(\lambda) \quad (w \geq 1). \quad (1)$$

For a finite sequence $a = (a_1, \dots, a_m)$ of non-negative integers let $X_a = \{(i, j) : 1 \leq j \leq m, 1 \leq i \leq a_j\}$. If a is non-increasing, then $X_a = Y(\nu)$ where ν is the partition formed by the non-zero entries of a .

For $w \geq 1$ the w -th diagonal is $D_w = \{(i, j) \in \mathbb{P} : i + j = w + 1\}$; it has the w cells $d_w(j) = (w + 1 - j, j)$, $1 \leq j \leq w$ ($d_w(j)$ is the cell of D_w in column j). $\mathbb{P}$ is the disjoint union of the D_w . Put $\Delta_0 = \emptyset$ and $\Delta_m = D_1 \cup \dots \cup D_m$ for $m \geq 1$. Column j of Δ_m consists of the cells (i, j) with $i \leq m + 1 - j$, so $\Delta_m = Y(\delta_m)$ with $\delta_m = (m, m - 1, \dots, 1)$, and $|\Delta_m| = T_m$.

Define $\rho : \mathbb{P} \rightarrow \mathbb{P}$ by $\rho(i, j) = (i - 1, j + 1)$ if $i \geq 2$ and $\rho(1, j) = (j, 1)$.

Lemma 2.1. (a) $\rho(d_w(j)) = d_w(j + 1)$ for $1 \leq j < w$ and $\rho(d_w(w)) = d_w(1)$. Hence ρ maps each D_w bijectively onto itself, ρ is a bijection of $\mathbb{P}$, and for $m \geq 0$, $\rho^m(d_w(j)) = d_w(j')$ where $1 \leq j' \leq w$ and $j' \equiv j + m \pmod{w}$. In particular ρ^w is the identity on D_w . (b) ρ preserves the quantity $i + j$ of a cell.

Proof. (a) For $j < w$ the cell $d_w(j) = (w + 1 - j, j)$ has row $w + 1 - j \geq 2$, so $\rho(d_w(j)) = (w - j, j + 1) = d_w(j + 1)$; and $\rho(d_w(w)) = \rho(1, w) = (w, 1) = d_w(1)$. So ρ acts on D_w as the cyclic shift $j \mapsto j + 1$ of column indices modulo w ; the rest follows. (b) $(i - 1) + (j + 1) = i + j$ and $j + 1 = 1 + j$. $\square$

Lemma 2.2 (one move). Let $\lambda \vdash n$, $s = \ell(\lambda)$ and $a = (s, \lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_s - 1)$. Then $\rho(Y(\lambda)) = X_a$, and $Y(\mathcal{B}(\lambda)) = X_{a^*}$, where a^* is the non-increasing rearrangement of a . If $s \geq \lambda_1 - 1$, then a is non-increasing and $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.

Proof. The cells of row 1 of $Y(\lambda)$ are $(1, j)$, $1 \leq j \leq s$; ρ sends them to $(j, 1)$, i.e. exactly onto the cells $(1, 1), \dots, (s, 1)$ of column 1. A cell $(i, j) \in Y(\lambda)$ with $i \geq 2$ satisfies $2 \leq i \leq \lambda_j$ and is sent to $(i - 1, j + 1)$; for fixed j these images are exactly the cells $(1, j + 1), \dots, (\lambda_j - 1, j + 1)$. Hence $\rho(Y(\lambda)) = X_a$. The parts of $\mathcal{B}(\lambda)$ are the non-zero entries of a , so $Y(\mathcal{B}(\lambda)) = X_{a^*}$. The entries of a after the first are non-increasing because λ is; so a is non-increasing iff $s \geq \lambda_1 - 1$, and then $a^* = a$. $\square$

For a finite set $X \subseteq \mathbb{P}$ put $\Phi(X) = \sum_{(i, j) \in X} (i + j)$.

Lemma 2.3. For every finite sequence a of non-negative integers, $\Phi(X_{a^*}) \leq \Phi(X_a)$, with equality iff a is non-increasing.

Proof. $\Phi(X_a) = \sum_j \sum_{i=1}^{a_j} (i + j) = \sum_j \binom{a_j + 1}{2} + F(a)$ with $F(a) = \sum_j j a_j$. The first sum does not change when a is permuted. If $a_j < a_{j+1}$, exchanging these two entries changes F by $j a_{j+1} + (j + 1) a_j - j a_j - (j + 1) a_{j+1} = a_j - a_{j+1} < 0$. Starting from a , repeatedly exchange some adjacent pair with $a_j < a_{j+1}$. Each exchange lowers the non-negative integer F , so the process stops, and it stops at a sequence with no adjacent ascent, i.e. at the non-increasing rearrangement a^* (which is unique). If a is not non-increasing at least one exchange happens, so $F(a^*) < F(a)$; if a is non-increasing then $a = a^*$. $\square$

Corollary 2.4. For $\lambda \vdash n$: $\Phi(Y(\mathcal{B}(\lambda))) \leq \Phi(Y(\lambda))$, with equality iff $\ell(\lambda) \geq \lambda_1 - 1$; in the case of equality $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.

Proof. By Lemma 2.1, ρ is injective and preserves $i + j$, so $\Phi(\rho(X)) = \Phi(X)$. With a as in Lemma 2.2: $\Phi(Y(\lambda)) = \Phi(X_a) \geq \Phi(X_{a^*}) = \Phi(Y(\mathcal{B}(\lambda)))$, with equality iff a is non-increasing (Lemma 2.3), i.e. iff $\ell(\lambda) \geq \lambda_1 - 1$, and then Lemma 2.2 gives $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$. $\square$

For $k \geq 1$ and $E \subseteq D_k$ let λ_E be the partition whose j -th part, $1 \leq j \leq k$, is $k - j + [d_k(j) \in E]$ (zero parts omitted). The sequence $(k - j + \varepsilon_j)_{j=1}^k$ with $\varepsilon_j \in \{0, 1\}$ is non-increasing (consecutive differences are $1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$), so λ_E is a partition, $|\lambda_E| = T_{k-1} + |E|$, and $Y(\lambda_E) = \Delta_{k-1} \cup E$ (column j consists of the $k - j$ cells of Δ_{k-1} and, if $d_k(j) \in E$, the cell $d_k(j) = (k + 1 - j, j)$). Writing $e_{k-j} = [d_k(j) \in E]$ we get $\lambda_E = (k - 1 + e_{k-1}, k - 2 + e_{k-2}, \dots, 1 + e_1, e_0)$.

Theorem 2.5 (Brandt [2]; see also [1, 3, 4]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition of n is cyclic if and only if it equals λ_E for some $E \subseteq D_k$ with $|E| = r$; equivalently, iff it has the form $(k - 1 + e_{k-1}, \dots, 1 + e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted). Moreover $Y(\mathcal{B}(\lambda_E)) = \rho(Y(\lambda_E))$, i.e. $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and $\mathcal{B}^k(\lambda_E) = \lambda_E$.*

Proof. ($\Leftarrow$ and “moreover”). Let $\mu = \lambda_E$ with $E \subseteq D_k$, $|E| = r$. Row 1 of $Y(\mu) = \Delta_{k-1} \cup E$ consists of the cells $(1, w) = d_w(w)$ with $w \leq k - 1$ and possibly $d_k(k)$; so $\ell(\mu) = k - 1 + [d_k(k) \in E] \geq k - 1$. Column 1 consists of $d_w(1)$ for $w \leq k - 1$ and possibly $d_k(1)$; so $\mu_1 = k - 1 + [d_k(1) \in E] \leq k$. Thus $\ell(\mu) \geq \mu_1 - 1$ and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(\Delta_{k-1} \cup \rho(E)) = \Delta_{k-1} \cup \rho(E)$ with $\rho(E) \subseteq D_k$, $|\rho(E)| = r$. Hence $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and by induction $\mathcal{B}^k(\lambda_E) = \lambda_{\rho^k(E)} = \lambda_E$ because ρ^k is the identity on D_k . So μ is cyclic.

($\Rightarrow$). Let $\lambda \vdash n$ with $\mathcal{B}^P(\lambda) = \lambda$, $P \geq 1$, and write $\lambda^{(m)} = \mathcal{B}^m(\lambda)$. By Corollary 2.4, $\Phi(Y(\lambda^{(0)})) \geq \Phi(Y(\lambda^{(1)})) \geq \dots \geq \Phi(Y(\lambda^{(P)})) = \Phi(Y(\lambda^{(0)}))$, so all these are equalities and $Y(\lambda^{(m+1)}) = \rho(Y(\lambda^{(m)}))$ for $0 \leq m < P$. As $\lambda^{(m+P)} = \lambda^{(m)}$, this holds for all $m \geq 0$, and therefore $Y(\lambda^{(m)}) = \rho^m(Y(\lambda))$ for all $m \geq 0$.

Claim: there is no $w \geq 1$ such that $D_w \not\subseteq Y(\lambda)$ and $D_{w+1} \cap Y(\lambda) \neq \emptyset$. Suppose $d_w(\alpha) \notin Y(\lambda)$ and $d_{w+1}(\beta) \in Y(\lambda)$. Since $\gcd(w, w+1) = 1$, the Chinese remainder theorem gives $m \geq 0$ with $m \equiv w - \alpha \pmod{w}$ and $m \equiv w + 1 - \beta \pmod{w+1}$. By Lemma 2.1, $\rho^m(d_w(\alpha)) = d_w(w) = (1, w)$ and $\rho^m(d_{w+1}(\beta)) = d_{w+1}(w+1) = (1, w+1)$. As ρ^m is injective, $(1, w) \notin \rho^m(Y(\lambda)) = Y(\lambda^{(m)})$ while $(1, w+1) \in Y(\lambda^{(m)})$, contradicting (1).

$Y(\lambda)$ is finite, so some D_w is not contained in it; let $K \geq 1$ be the least such w . Then $\Delta_{K-1} \subseteq Y(\lambda)$. By the claim, $D_{K+1} \cap Y(\lambda) = \emptyset$; then $D_{K+1} \not\subseteq Y(\lambda)$, so again by the claim $D_{K+2} \cap Y(\lambda) = \emptyset$, and inductively $D_w \cap Y(\lambda) = \emptyset$ for all $w > K$. Thus $Y(\lambda) = \Delta_{K-1} \cup E'$ with $E' = Y(\lambda) \cap D_K \neq D_K$; let $r' = |E'| \in \{0, \dots, K-1\}$, so $n = T_{K-1} + r'$ and $Y(\lambda) = Y(\lambda_{E'})$ (with k replaced by K in the definition of $\lambda_{E'}$). If $r' \geq 1$, then $T_{K-1} < n < T_K$, so by Lemma 1.2 $K = k$, $r' = r$, and $\lambda = \lambda_{E'}$. If $r' = 0$, then $K \geq 2$ (as $n \geq 1$) and $n = T_{K-1} = T_{K-2} + (K-1)$, so $k = K-1$, $r = k$, and $Y(\lambda) = \Delta_k = \Delta_{k-1} \cup D_k$, i.e. $\lambda = \lambda_{D_k}$ with $|D_k| = k = r$. $\square$

Corollary 2.6. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$.*

- (a) *If $r = k$ (i.e. $n = T_k$), the only cyclic partition of n is $\delta_k = (k, k-1, \dots, 1)$, and $\mathcal{B}(\delta_k) = \delta_k$.*
- (b) *Every cyclic $\mu \vdash n$ satisfies $Y(\mu) \subseteq \Delta_k$, $\Delta_{k-1} \subseteq Y(\mu)$, $\ell(\mu) \in \{k-1, k\}$ and $\mathcal{B}^k(\mu) = \mu$.*
- (c) *If $\mu \vdash n$ is cyclic, then for every $a \geq 0$: $\sum_{i=a}^{a+k-1} \ell(\mathcal{B}^i(\mu)) = k(k-1) + r$. If moreover $r < k$, both values $k-1$ and k occur among $\ell(\mathcal{B}^a(\mu)), \dots, \ell(\mathcal{B}^{a+k-1}(\mu))$.*

Proof. (a) and (b) are immediate from Theorem 2.5 ($E = D_k$ when $r = k$; $\ell(\lambda_E) = k-1 + [d_k(k) \in E]$ was computed in its proof). (c) Write $\mu = \lambda_E$. Then $\mathcal{B}^i(\mu) = \lambda_{\rho^i(E)}$, so $\ell(\mathcal{B}^i(\mu)) = k-1 + [d_k(k) \in \rho^i(E)] = k-1 + [d_k(j_i) \in E]$, where $d_k(j_i) = \rho^{-i}(d_k(k))$, i.e. $j_i \equiv k-i \pmod{k}$ (Lemma 2.1). As i runs through $a, \dots, a+k-1$, j_i runs through all of $1, \dots, k$ once, so the sum is $k(k-1) + |E| = k(k-1) + r$. If $0 < r < k$, some j_i lies in E and some does not. $\square$

3 Piles and the part-count sequence

This is the “ $\text{diagram}_B(\lambda)$ ” bookkeeping of Griggs–Ho [4, §3], written with explicitly labelled piles.

Fix $\lambda \vdash n$, write $\lambda^{(i)} = \mathcal{B}^i(\lambda)$ and

$$c_i = \ell(\lambda^{(i-1)}) \quad (i \geq 1).$$

We call $c(\lambda) = (c_1, c_2, \dots)$ the *part-count sequence*; clearly $c(\lambda^{(a)}) = (c_{a+1}, c_{a+2}, \dots)$. Every $c_i \geq 1$.

Lemma 3.1 (piles). *For every $i \geq 0$ the multiset of parts of $\lambda^{(i)}$ is*

$$\{\lambda_m - i : 1 \leq m \leq \ell(\lambda), \lambda_m > i\} \uplus \{c_j - (i - j) : 1 \leq j \leq i, c_j > i - j\}.$$

Proof. Induction on i ; $i = 0$ is clear. $\lambda^{(i+1)} = \mathcal{B}(\lambda^{(i)})$ has the part $\ell(\lambda^{(i)}) = c_{i+1} = c_{i+1} - 0$ and the parts of $\lambda^{(i)}$ decreased by one, the zeros being discarded: this is the stated multiset for $i + 1$ (the term $j = i + 1$ is c_{i+1}). $\square$

We use *labels*: $O_1, \dots, O_{\ell(\lambda)}$ (the original piles) and $N_1, N_2, \dots$ (N_j is the pile created by the j -th move, of size c_j when created). By Lemma 3.1, at time i (i.e. in $\lambda^{(i)}$) the pile O_m has size $\lambda_m - i$ and the pile N_j ($j \leq i$) has size $c_j - (i - j)$, a pile being present iff its size is positive. For $i \geq 1$ let R_i be the set of labels of the piles present at time $i - 1$:

$$O_m \in R_i \iff \lambda_m \geq i, \quad N_j \in R_i \iff j \leq i - 1 \text{ and } c_j \geq i - j. \quad (2)$$

By Lemma 3.1 (with $i - 1$), $|R_i| = \ell(\lambda^{(i-1)}) = c_i$. Let $Z_i = R_i \setminus R_{i+1}$ (the piles of size 1 at time $i - 1$, which disappear in move i).

Lemma 3.2 (basic facts). *For all $i \geq 1$:*

- (a) $N_i \in R_{i+1}$ and $N_i \notin R_i$.
- (b) $R_{i+1} = (R_i \setminus Z_i) \cup \{N_i\}$, hence $c_{i+1} = c_i + 1 - |Z_i|$ and $c_{i+1} \leq c_i + 1$.
- (c) If $c_{i+1} = c_i + 1$ then $Z_i = \emptyset$, i.e. $R_i \subseteq R_{i+1}$. If $c_{i+1} = c_i$ then $|Z_i| = 1$.
- (d) For $j < i$: $N_j \in R_i$ implies $N_j \in R_{i'}$ for $j < i' \leq i$, and $N_j \notin R_i$ implies $N_j \notin R_{i'}$ for all $i' \geq i$. Likewise for the labels O_m .

Proof. (a) $c_i \geq 1 = (i + 1) - i$; and labels N_j in R_i have $j \leq i - 1$. (d) For $i' > j$ the clause $j \leq i' - 1$ in (2) holds automatically, so membership of N_j in $R_{i'}$ is the condition $c_j \geq i' - j$, which is monotone in i' ; for O_m the condition $\lambda_m \geq i'$ is monotone as well. (b) By (d), $R_{i+1} \setminus \{N_i\} \subseteq R_i$, so $R_{i+1} = (R_i \cap R_{i+1}) \cup \{N_i\} = (R_i \setminus Z_i) \cup \{N_i\}$, a disjoint union by (a); take cardinalities. (c) follows from $c_{i+1} = c_i + 1 - |Z_i|$. $\square$

Lemma 3.3 (gap lemma). *Let $1 \leq j < i$. If $N_j \notin R_i$ and $N_{j+1} \in R_{i+1}$, then $c_j = i - j - 1$, $c_{j+1} = i - j$, and $N_{j+1} \notin R_{i+2}$.*

Proof. By (2), $c_j \leq i - j - 1$ and $c_{j+1} \geq (i + 1) - (j + 1) = i - j$. By Lemma 3.2(b), $c_{j+1} \leq c_j + 1 \leq i - j$. Hence both equalities hold, and $c_{j+1} = i - j < (i + 2) - (j + 1)$ gives $N_{j+1} \notin R_{i+2}$. $\square$

Lemma 3.4 (the sequence determines the partition). *For $i \geq 1$, $\#\{m : \lambda_m \geq i\} = c_i - \#\{j : 1 \leq j \leq i - 1, c_j \geq i - j\}$. Consequently, if two partitions have the same part-count sequence, they are equal.*

Proof. The formula is $|R_i| = c_i$ together with (2). The numbers $\#\{m : \lambda_m \geq i\}$, $i \geq 1$, determine λ , since $\lambda_m = \#\{i \geq 1 : \#\{m' : \lambda_{m'} \geq i\} \geq m\}$. $\square$

Lemma 3.5 (window sums; generalising [4, Prop. 3.2(2)]). *For all $a \geq 0$ and $h \geq 1$: $c_{a+1} + c_{a+2} + \dots + c_{a+h} \leq n + T_{h-1}$.*

Proof. Let $\mu = \lambda^{(a)}$; its part-count sequence is $(c_{a+1}, c_{a+2}, \dots)$. Applying (2) to μ , for $1 \leq g \leq h$ the number c_{a+g} equals $\#\{m : \mu_m \geq g\}$ plus the number of piles created in the first $g-1$ moves starting from μ that are still present, which is at most $g-1$. Summing, $\sum_{g=1}^h c_{a+g} \leq \sum_{g \geq 1} \#\{m : \mu_m \geq g\} + \sum_{g=1}^h (g-1) = |\mu| + T_{h-1} = n + T_{h-1}$. $\square$

4 Ramps

The following lemmas are Lemmas 3.4–3.6 and Proposition 3.2(3) of Griggs–Ho [4]; we give complete proofs (the proof of [4, Lemma 3.5] ends with “continue this process”, which we replace by an explicit induction) and we add the equality statement in Proposition 4.5.

Definition 4.1. A *ramp* of λ is a triple (p, q, x) of integers with $p \geq 1$, $q \geq p+2$ and

$$c_p = x-1, \quad c_{p+1} = c_{p+2} = \dots = c_{q-1} = x, \quad c_q = x+1.$$

Its *length* is $q-p$ and its *value* is x (note $x \geq 2$, since $c_p \geq 1$).

Lemma 4.2 (sandwich). *If $1 \leq i < j$ and $c_i < x < c_j$ for an integer x , then there is a ramp (p, q, x) with $i \leq p < q \leq j$.*

Proof. Let p be the largest m with $i \leq m < j$ and $c_m \leq x-1$ (it exists: $m=i$ qualifies). By Lemma 3.2(b), $c_{p+1} \leq c_p + 1 \leq x$. Since $c_j > x$, this rules out $p+1 = j$; so $p+1 < j$, and $c_{p+1} \geq x$ by maximality of p . Hence $c_p = x-1$ and $c_{p+1} = x$. Let q be the least m with $p < m \leq j$ and $c_m \geq x+1$ (it exists: $m=j$ qualifies). Then $q \geq p+2$ since $c_{p+1} = x$. For $p < m < q$ we have $c_m \leq x$ (minimality of q) and $c_m \geq x$ (maximality of p , as $m < j$), so $c_m = x$. Finally $x+1 \leq c_q \leq c_{q-1} + 1 = x+1$. $\square$

Lemma 4.3 (short ramps [4, Lemma 3.6]). *If $(p, p+2, x)$ is a ramp, then $p \leq x$.*

Proof. Suppose $p \geq x+1$; then $p \geq 3$. Since $|R_p| = c_p = x-1 \leq p-2$, not all of $N_1, \dots, N_{p-1}$ lie in R_p , while $N_{p-1} \in R_p$ (Lemma 3.2(a)). Let j be the largest index in $[1, p-2]$ with $N_j \notin R_p$; then $N_{j+1} \in R_p$. As $c_{p+1} = c_p + 1$ and $c_{p+2} = c_{p+1} + 1$, Lemma 3.2(c) gives $R_p \subseteq R_{p+1} \subseteq R_{p+2}$, so $N_{j+1} \in R_{p+1}$ and $N_{j+1} \in R_{p+2}$. But Lemma 3.3 (with $i=p$) gives $N_{j+1} \notin R_{p+2}$, a contradiction. $\square$

Lemma 4.4 (descent [4, Lemma 3.5]). *Let (p, q, x) be a ramp with $q-p \geq 3$ and $p \geq x+1$. Then there is a ramp (p', q', x') with*

$$p-x \leq p' \leq p-2, \quad x' = p-p' \leq x, \quad 2 \leq q'-p' \leq q-p-1.$$

Proof. The x labels $N_{p-x}, \dots, N_{p-1}$ exist since $p-x \geq 1$. As $|R_p| = x-1$, one of them is not in R_p ; $N_{p-1} \in R_p$. Let p' be the largest index in $[p-x, p-2]$ with $N_{p'} \notin R_p$ and put $y = p-p'$, so $2 \leq y \leq x$ and $N_{p'+1}, \dots, N_{p-1} \in R_p$. As $c_{p+1} = c_p + 1$, $R_p \subseteq R_{p+1}$, and with Lemma 3.2(a), $N_{p'+1}, \dots, N_p \in R_{p+1}$. Lemma 3.3 ($j=p'$, $i=p$) gives $c_{p'} = y-1$ and $c_{p'+1} = y$.

For $1 \leq l \leq q-p-1$ let $H(l)$ be the statement

$$H(l) : \quad c_{p'+1} = \dots = c_{p'+l} = y \quad \text{and} \quad N_{p'+l}, N_{p'+l+1}, \dots, N_{p+l-1} \in R_{p+l}.$$

We just proved $H(1)$. Assume $H(l)$. Since $c_{p'+l} = y = p-p'$, (2) gives, for $i > p'+l$: $N_{p'+l} \in R_i \iff i \leq p+l$. Hence $N_{p'+l} \in Z_{p+l}$.

- If $l = q - p - 1$, then $p + l = q - 1$ and $c_q = c_{q-1} + 1$, so $Z_{q-1} = \emptyset$ by Lemma 3.2(c): a contradiction. So $H(q - p - 1)$ is false.
- If $l \leq q - p - 2$, then $p + l$ and $p + l + 1$ lie in $[p + 1, q - 1]$, so $c_{p+l+1} = c_{p+l} = x$ and $|Z_{p+l}| = 1$, i.e. $Z_{p+l} = \{N_{p'+l}\}$. By Lemma 3.2(b), $R_{p+l+1} = (R_{p+l} \setminus \{N_{p'+l}\}) \cup \{N_{p+l}\} \supseteq \{N_{p'+l+1}, \dots, N_{p+l}\}$. In particular $N_{p'+l+1} \in R_{p+l+1}$, i.e. $c_{p'+l+1} \geq (p + l + 1) - (p' + l + 1) = y$, and $c_{p'+l+1} \leq c_{p+l} + 1 = y + 1$. So either $c_{p'+l+1} = y + 1$, or $c_{p'+l+1} = y$ and then $H(l + 1)$ holds.

Let l_0 be the largest $l \leq q - p - 1$ such that $H(1), \dots, H(l)$ all hold; then $1 \leq l_0 \leq q - p - 2$, and since $H(l_0 + 1)$ fails, the second item gives $c_{p'+l_0+1} = y + 1$. Thus $(p', p' + l_0 + 1, y)$ is a ramp of length $l_0 + 1 \in [2, q - p - 1]$, and $x' = y$ has the stated properties. $\square$

Proposition 4.5 (ramp bound). *Every ramp (p, q, x) satisfies $p \leq (q - p - 1)x$. If equality holds, then $(jx, jx + j + 1, x)$ is a ramp for every $j = 1, 2, \dots, q - p - 1$.*

Proof. Strong induction on the length $L = q - p \geq 2$. If $L = 2$, Lemma 4.3 gives $p \leq x$; if $p = x$, the ramp itself is $(x, x + 2, x)$. Let $L \geq 3$. If $p \leq x$ then $p \leq x < (L - 1)x$. Otherwise Lemma 4.4 provides a ramp (p', q', x') with $p' \geq p - x$, $x' \leq x$ and length $L' \leq L - 1$; by induction $p' \leq (L' - 1)x'$, hence

$$p \leq p' + x \leq (L' - 1)x' + x \leq (L - 2)x + x = (L - 1)x.$$

If $p = (L - 1)x$, all these inequalities are equalities: $p' = p - x = (L - 2)x$ and $(L' - 1)x' = (L - 2)x$; since $1 \leq L' - 1 \leq L - 2$ and $x' \leq x$, this forces $L' = L - 1$ and $x' = x$. So $((L - 2)x, (L - 2)x + L - 1, x)$ is a ramp attaining equality, and by induction $(jx, jx + j + 1, x)$ is a ramp for $j = 1, \dots, L - 2$; for $j = L - 1$ it is (p, q, x) itself. $\square$

Lemma 4.6 (long ramps [4, Lemma 3.4]). *If (p, q, x) is a ramp with $q - p = x + 1$, then $q \leq n + 1$. (Recall that $x \geq 2$ for every ramp.)*

Proof. For $p + 1 \leq j \leq q - 1$ we have $c_j = x \geq q - j$, so $N_j \in R_q$: these are x piles. Also $c_p = x - 1 < q - p$, so $N_p \notin R_q$. Since $|R_q| = c_q = x + 1$, R_q contains exactly one further label P , and $P = O_m$ for some m , or $P = N_j$ with $1 \leq j \leq p - 1$. If $P = O_m$, then $q \leq \lambda_m \leq n$. If $P = N_1$, then $c_1 \geq q - 1$, and $c_1 = \ell(\lambda) \leq n$. Suppose $P = N_j$ with $2 \leq j \leq p - 1$. Then $N_{j-1} \notin R_q$. As $c_q = c_{q-1} + 1$, $R_{q-1} \subseteq R_q$, so $N_{j-1} \notin R_{q-1}$. Next, $q - 2 \geq p + 1$ (as $x \geq 2$), so $c_{q-1} = c_{q-2} = x$ and $|Z_{q-2}| = 1$; since $c_p = x - 1 = (q - 2) - p$, $N_p \in R_{q-2} \setminus R_{q-1}$, so $Z_{q-2} = \{N_p\}$. As $j - 1 \neq p$, $N_{j-1} \in R_{q-2}$ would give $N_{j-1} \in R_{q-1}$ (Lemma 3.2(b)); hence $N_{j-1} \notin R_{q-2}$, i.e. $c_{j-1} \leq q - j - 2$. But $N_j \in R_q$ gives $c_j \geq q - j$, so $c_j \geq c_{j-1} + 2$, contradicting Lemma 3.2(b). $\square$

5 The cyclic partitions of $T_{k-1} + 2$ and how orbits enter them

Let $k \geq 5$ and $n = T_{k-1} + 2$ (rank k , $r = 2$). For $j \in \{1, \dots, k\}$ put $j \oplus 1 = j + 1$ for $j < k$, $k \oplus 1 = 1$, and $j \ominus 1$ for the inverse operation. For $E \subseteq \{1, \dots, k\}$ with $|E| = 2$ let $\zeta_E = \lambda_{\{d_k(j): j \in E\}}$. Its i -th part is $p_i = k - i + [i \in E]$ ($1 \leq i \leq k$, zero parts omitted), so ζ_E is δ_{k-1} with 1 added to its parts number $j \in E$. By Theorem 2.5 and Lemma 2.1:

$$\begin{aligned} \text{the cyclic partitions of } n \text{ are exactly the } \zeta_E, |E| = 2; \quad \mathcal{B}(\zeta_E) = \zeta_{E \oplus 1}; \\ \ell(\zeta_E) = k - 1 + [k \in E], \end{aligned} \tag{3}$$

where $E \oplus 1 = \{j \oplus 1 : j \in E\}$. In particular $\ell(\zeta_{E \oplus 1}) = k - 1 + [1 \in E]$. Note that $p_1 = k - 1 + [1 \in E]$ is the largest part, $p_2 = k - 2 + [2 \in E]$, and $p_i \leq k - 2$ for $i \geq 3$, with $p_3 = k - 2$ iff $3 \in E$.

Lemma 5.1 (preimages). *Let ζ be a partition and $s \geq 1$. There is at most one partition μ with $\ell(\mu) = s$ and $\mathcal{B}(\mu) = \zeta$. It exists iff s is a part of ζ and $s \geq \ell(\zeta) - 1$. Its parts are then the numbers $z + 1$ for z in the multiset $\zeta \ominus \{s\}$ (ζ with one copy of s removed), together with $s - \ell(\zeta) + 1$ parts equal to 1.*

Proof. If $\mathcal{B}(\mu) = \zeta$ and $\ell(\mu) = s$, then by definition the parts of ζ are s and the $\mu_m - 1$ with $\mu_m \geq 2$. So s is a part of ζ , and $\{\mu_m - 1 : \mu_m \geq 2\} = \zeta \ominus \{s\}$ has $\ell(\zeta) - 1$ elements. The remaining $s - \ell(\zeta) + 1 \geq 0$ parts of μ equal 1. This determines μ . Conversely, the μ so described has s parts and $\mathcal{B}(\mu) = \zeta$. $\square$

Lemma 5.2 (entry lemma for $r = 2$). *Let $k \geq 5$, $n = T_{k-1} + 2$, $\lambda \vdash n$, $d = d_{\mathcal{B}}(\lambda) \geq 1$, $\mu = \lambda^{(d-1)}$, and write $\lambda^{(d)} = \zeta_E$. Then $k \notin E$, and:*

- (a) *if $E \neq \{1, 2\}$, then $\ell(\mu) = k - 2$, and E is not of the form $\{2, b\}$ with $b \geq 4$; moreover $\mu = (k + 1, k - 1, k - 3, k - 4, \dots, 2)$ if $E = \{1, 3\}$, and $\mu = (k, k, k - 3, k - 4, \dots, 2)$ if $E = \{2, 3\}$;*
- (b) *if $E = \{1, 2\}$, then $\mu = (k + 1, k - 2, k - 3, \dots, 2, 1)$, which has $k - 1$ parts.*

(Here “ $k - 3, k - 4, \dots, 2$ ” denotes the decreasing run from $k - 3$ down to 2.)

Proof. μ is not cyclic and $\mathcal{B}(\mu) = \zeta_E$. Let $s = \ell(\mu)$. By Lemma 5.1, s is a part of ζ_E , $s \geq \ell(\zeta_E) - 1$, and μ is determined by s . The cyclic partition $\zeta_{E \oplus 1}$ also satisfies $\mathcal{B}(\zeta_{E \oplus 1}) = \zeta_E$ and differs from μ , so $s \neq \ell(\zeta_{E \oplus 1}) = k - 1 + [1 \in E]$.

$k \in E$ is impossible. Then $\ell(\zeta_E) = k$, so $k - 1 \leq s \leq p_1 = k - 1 + [1 \in E]$. If $1 \notin E$, then $s = k - 1 = \ell(\zeta_{E \oplus 1})$, which is excluded. If $1 \in E$, i.e. $E = \{1, k\}$, then $s = k = \ell(\zeta_{E \oplus 1})$ is excluded. And $s = k - 1$ is not a part of $\zeta_{\{1, k\}}$, whose parts are $k, k - 2$ and numbers $\leq k - 3$.

$k \notin E$. Then $\ell(\zeta_E) = k - 1$, so $s \geq k - 2$, and $s \leq p_1 \leq k$.

- $1 \notin E$: $s \neq k - 1$ and $s \leq p_1 = k - 1$, so $s = k - 2$. In particular $k - 2$ must be a part, i.e. $2 \notin E$ or $3 \in E$; this excludes $E = \{2, b\}$ with $b \geq 4$.
- $1 \in E$, $2 \notin E$: $s \neq k$, and $k - 1$ is not a part ($p_1 = k$, $p_2 = k - 2$, $p_i \leq k - 2$ for $i \geq 3$), so $s = k - 2$.
- $E = \{1, 2\}$: $s \neq k$. The parts are $k, k - 1, k - 3, k - 4, \dots, 1$ ($p_3 = k - 3$ as $3 \notin E$), so $k - 2$ is not a part and $s = k - 1$.

Now Lemma 5.1 gives μ . For $E = \{1, 2\}$: $\zeta_E \ominus \{k - 1\} = \{k, k - 3, k - 4, \dots, 1\}$, plus one part 1, so $\mu = (k + 1, k - 2, \dots, 2, 1)$. For $E = \{1, 3\}$: $\zeta_E = (k, k - 2, k - 2, k - 4, \dots, 1)$ and $\zeta_E \ominus \{k - 2\} = \{k, k - 2, k - 4, \dots, 1\}$, with no parts 1 added, so $\mu = (k + 1, k - 1, k - 3, \dots, 2)$. For $E = \{2, 3\}$: $\zeta_E = (k - 1, k - 1, k - 2, k - 4, \dots, 1)$, so $\mu = (k, k, k - 3, \dots, 2)$. $\square$

6 The pile-chain lemma

For a label P present at time T (i.e. contributing a part of $\lambda^{(T)}$), write $|P|_T$ for its size then. So $|N_j|_T = c_j - (T - j)$ ($j \leq T$) and $|O_m|_T = \lambda_m - T$.

Lemma 6.1 (pile chain). *Let $T \geq 1$.*

- (a) *If $2 \leq j \leq T$ and N_j is present at time T with $|N_j|_T = \sigma \geq 3$, then N_{j-1} is present at time T and $|N_{j-1}|_T \geq \sigma - 2$.*

(b) If $P \in \{N_1\} \cup \{O_m\}$ is present at time T with $|P|_T = \sigma$, then $T \leq n - \sigma + 1$.

Proof. (a) By Lemma 3.2(b), $c_j \leq c_{j-1} + 1$. So $c_{j-1} - (T - j + 1) \geq c_j - (T - j) - 2 = \sigma - 2 \geq 1$, and N_{j-1} is present with that size. (b) If $P = N_1$, then $\sigma = c_1 - (T - 1)$ and $c_1 = \ell(\lambda) \leq n$. If $P = O_m$, then $\sigma = \lambda_m - T$ and $\lambda_m \leq n$. $\square$

In words: a pile of size σ that is not among the first ones forces an *older* pile of size at least $\sigma - 2$ to be present at the same time. So in a partition with a single very large part, that part must come from the first move (or from λ itself), and it has been shrinking since time 1.

7 The upper bound

Theorem 7.1. *Let $k \geq 7$ and $\lambda \vdash T_{k-1} + 2$. Then $d_{\mathcal{B}}(\lambda) \leq (k-1)(k-4)$.*

Proof. Let $n = T_{k-1} + 2$ and $d = d_{\mathcal{B}}(\lambda)$. If $d \leq 1$ there is nothing to prove, so assume $d \geq 2$ (hence $T = d - 1 \geq 1$). Let $\mu = \lambda^{(d-1)}$, $T = d - 1$ and $\lambda^{(d)} = \zeta_E$. By Lemma 5.2, $k \notin E$. We first bound d in each case, then compare with $(k-1)(k-4)$.

Case A: $E \neq \{1, 2\}$. Then $c_d = \ell(\mu) = k - 2$. Let $b = \max E \leq k - 1$. For $m \geq 0$, $c_{d+1+m} = \ell(\zeta_{E \oplus m}) = k - 1 + [k \in E \oplus m]$ by (3), and the least m with $k \in E \oplus m$ is $m = k - b$. So $(d, d + 1 + k - b, k - 1)$ is a ramp of length $k - b + 1 \in [2, k - 1]$, and Proposition 4.5 gives

$$d \leq (k - b)(k - 1). \quad (4)$$

If $b \geq 4$, this is at most $(k - 4)(k - 1)$. Otherwise $b \leq 3$, so $E = \{1, 3\}$ or $E = \{2, 3\}$, and we argue separately.

Case A1: $E = \{2, 3\}$, $\mu = (k, k, k - 3, \dots, 2)$. Among the two piles of size k present at time T , let P be the older one: an O_m if possible, otherwise the one N_j with the smaller j . If $P = N_j$ with $j \geq 2$, then Lemma 6.1(a) gives a pile N_{j-1} of size $\geq k - 2$. The only parts of μ that are $\geq k - 2$ are the two k 's, so N_{j-1} would be the other k -pile, which is older than P . This is a contradiction. So $P \in \{N_1\} \cup \{O_m\}$, and Lemma 6.1(b) gives $T \leq n - k + 1$, i.e. $d \leq n - k + 2$.

Case A2: $E = \{1, 3\}$, $\mu = (k + 1, k - 1, k - 3, \dots, 2)$. The parts of μ are one $k + 1$, one $k - 1$, and numbers $\leq k - 3$. If $d \leq 3$ we are done, so let $d \geq 4$. By Lemma 5.1 applied to $\mathcal{B}(\lambda^{(d-2)}) = \mu$, the number $c_{d-1} = \ell(\lambda^{(d-2)})$ is a part of μ and is $\geq \ell(\mu) - 1 = k - 3$; so $c_{d-1} \in \{k - 3, k - 1, k + 1\}$. The newest pile $N_T = N_{d-1}$ has size c_{d-1} at time T .

- $c_{d-1} = k - 3$: then $c_{d-1} = k - 3$, $c_d = k - 2$, $c_{d+1} = \ell(\lambda^{(d)}) = \ell(\zeta_{\{1,3\}}) = k - 1$, so $(d - 1, d + 1, k - 2)$ is a ramp of length 2. Lemma 4.3 gives $d - 1 \leq k - 2$.
- $c_{d-1} = k + 1$: then N_T is the pile of size $k + 1$. By Lemma 6.1(a), N_{T-1} is present with size $\geq k - 1$. As $N_{T-1} \neq N_T$, it is the unique pile of size $k - 1$, so $c_{d-2} - 1 = k - 1$. Again by Lemma 6.1(a) ($T - 1 \geq 2$), N_{T-2} is present with size $\geq k - 3$. All remaining parts are $\leq k - 3$, so its size is $k - 3$, i.e. $c_{d-3} - 2 = k - 3$. Thus $(c_{d-3}, c_{d-2}, c_{d-1}) = (k - 1, k, k + 1)$, $(d - 3, d - 1, k)$ is a ramp of length 2, and Lemma 4.3 gives $d - 3 \leq k$.
- $c_{d-1} = k - 1$: then N_T is the unique pile of size $k - 1$. Let P be the pile of size $k + 1$; $P \neq N_T$. If $P = N_j$ with $2 \leq j \leq T - 1$, Lemma 6.1(a) gives a pile $N_{j-1} \neq N_T$ of size $\geq k - 1$ other than P . But N_T is the only such pile, a contradiction. So $P \in \{N_1\} \cup \{O_m\}$, and Lemma 6.1(b) gives $T \leq n - k$, i.e. $d \leq n - k + 1$.

So in Case A2, $d \leq \max\{k-1, k+3, n-k+1\}$.

Case B: $E = \{1, 2\}$, $\mu = (k+1, k-2, \dots, 2, 1)$. Let P be the pile of size $k+1$. If $P = N_j$ with $j \geq 2$, Lemma 6.1(a) gives another pile of size $\geq k-1$, but all other parts of μ are $\leq k-2$. So $P \in \{N_1\} \cup \{O_m\}$, $T \leq n-k$ and $d \leq n-k+1$.

Comparison. With $n = T_{k-1} + 2$:

$$(k-1)(k-4) - (n-k+2) = \frac{1}{2}k(k-7) \geq 0, \quad (k-1)(k-4) - (k+3) = k^2 - 6k + 1 > 0 \quad (k \geq 7).$$

Hence in all cases $d \leq (k-1)(k-4)$. $\square$

The argument uses $k \geq 7$ only in the final comparison. For $k \leq 6$ the special entries win (see Theorem 9.1 and Section 11).

8 The lower bound: an explicit extremal family

For $k \geq 5$, $h \in \{1, \dots, k-1\}$ and $a \in \{1, \dots, k\}$ let (with $j \oplus_w 1 = j+1$ for $j < w$ and $w \oplus_w 1 = 1$)

$$Z_3(h, a) = \Delta_{k-2} \cup (D_{k-1} \setminus \{d_{k-1}(h)\}) \cup \{d_k(a), d_k(a \oplus_k 1), d_k(a \oplus_k 1 \oplus_k 1)\}.$$

Lemma 8.1. *Let $k \geq 5$ and let μ be a partition with $Y(\mu) = Z_3(h, a)$. If $(h, a) \neq (k-1, 1)$, then $Y(\mathcal{B}(\mu)) = Z_3(h \oplus_{k-1} 1, a \oplus_k 1)$. If $(h, a) = (k-1, 1)$, then $\mu = (k, k-1, k-2, k-4, k-5, \dots, 2)$ and $\mathcal{B}(\mu) = \zeta_{\{3,4\}}$, which is cyclic.*

Proof. Let $A = \{a, a \oplus_k 1, a \oplus_k 1 \oplus_k 1\}$, a set of three elements. Counting the cells $d_w(w)$ of row 1 and $d_w(1)$ of column 1: $\ell(\mu) = k-2 + [h \neq k-1] + [k \in A]$ and $\mu_1 = k-2 + [h \neq 1] + [1 \in A]$. So $\ell(\mu) \leq \mu_1 - 2$ iff $h = k-1$, $h \neq 1$, $k \notin A$ and $1 \in A$. Now $1 \in A$ iff $a \in \{k-1, k, 1\}$, and $k \notin A$ excludes $a \in \{k-2, k-1, k\}$. So this happens iff $(h, a) = (k-1, 1)$. Otherwise Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(Z_3(h, a)) = Z_3(h \oplus_{k-1} 1, a \oplus_k 1)$. If $(h, a) = (k-1, 1)$, the column lengths are:

- column 1: $(k-2)+1+1 = k$; column 2: $(k-3)+1+1 = k-1$; column 3: $(k-4)+1+1 = k-2$;
- column $j \in [4, k-2]$: $(k-1-j) + 1 = k-j$;
- columns $k-1$ and k : 0.

So $\mu = (k, k-1, k-2, k-4, \dots, 2)$ with $k-2$ parts. $\mathcal{B}(\mu)$ has the parts $k-2$ and $k-1, k-2, k-3, k-5, \dots, 1$, i.e. $\mathcal{B}(\mu) = (k-1, k-2, k-2, k-3, k-5, \dots, 1) = \zeta_{\{3,4\}}$. $\square$

Theorem 8.2. *For $k \geq 5$, $\mu_k = (k-2, k-2, k-3, \dots, 4, 3, 3, 2, 1)$ is a partition of $T_{k-1} + 2$ with $d_{\mathcal{B}}(\mu_k) = (k-1)(k-4)$. (Precisely: $(\mu_k)_1 = k-2$, $(\mu_k)_i = k-i$ for $2 \leq i \leq k-3$, $(\mu_k)_{k-2} = 3$, $(\mu_k)_{k-1} = 2$, $(\mu_k)_k = 1$; e.g. $(3, 3, 3, 2, 1)$ for $k=5$ and $(4, 4, 3, 3, 2, 1)$ for $k=6$.)*

Proof. The sum is $(k-2) + (T_{k-2} - 3) + 3 + 2 + 1 = T_{k-1} + 2$, since $(\mu_k)_2, \dots, (\mu_k)_{k-3}$ are $k-2, \dots, 3$. The diagram is $Z_3(1, k-2)$, i.e. the hole is $d_{k-1}(1)$ and $A = \{k-2, k-1, k\}$:

- column 1 has the $k-2$ cells of Δ_{k-2} ;
- column 2 has $(k-3) + 1$ ($2 \notin A$ as $k \geq 5$);
- column $j \in [3, k-3]$ has $(k-1-j) + 1 = k-j$;
- column $k-2$ has $1 + 1 + 1 = 3$, column $k-1$ has $0 + 1 + 1 = 2$, and column k has 1.

For $m \geq 0$ let $h_m \in [1, k-1]$ with $h_m \equiv 1+m \pmod{k-1}$ and $a_m \in [1, k]$ with $a_m \equiv k-2+m \pmod{k}$. Then $(h_m, a_m) = (k-1, 1)$ iff $m \equiv -1 \pmod{k-1}$ and $m \equiv 3 \pmod{k}$. $M = k^2 - 5k + 3$ satisfies both (modulo $k-1$, $k \equiv 1$ gives $M \equiv -1$). For $k \geq 5$, $0 \leq M < k(k-1)$, so M is the least non-negative solution by the Chinese remainder theorem. By Lemma 8.1 and induction, $Y(\mathcal{B}^m(\mu_k)) = Z_3(h_m, a_m)$ for $0 \leq m \leq M$, and $\mathcal{B}^{M+1}(\mu_k) = \zeta_{\{3,4\}}$ is cyclic. For $m \leq M$ the diagram misses a cell of D_{k-1} , while every cyclic partition of n contains Δ_{k-1} (Corollary 2.6(b)); so $\mathcal{B}^m(\mu_k)$ is not cyclic. Hence $d_{\mathcal{B}}(\mu_k) = M+1 = k^2 - 5k + 4 = (k-1)(k-4)$. $\square$

9 The value of $D_{\mathcal{B}}(T_{k-1} + 2)$

Theorem 9.1. (a) For every $k \geq 7$: $D_{\mathcal{B}}(T_{k-1} + 2) = (k-1)(k-4)$.

(b) For $k = 2, \dots, 6$ (i.e. $n = 3, 5, 8, 12, 17$): $D_{\mathcal{B}}(T_{k-1} + 2) = 2, 3, 5, 8, 12$ respectively. The extremal partitions are:

- $n = 3, 5, 8, 17$: $1^n = (1, 1, \dots, 1)$ only;
- $n = 12$: $(3, 3, 2, 2, 1, 1)$ and 1^{12} .

(c) The formula $(k-1)(k-4)$ holds exactly for $k \geq 7$. For all $k \geq 2$, $D_{\mathcal{B}}(T_{k-1} + 2) = \max\{(k-1)(k-4), T_{k-2} + 2\}$.

Proof. (a) Theorems 7.1 and 8.2. (b) Exhaustive computation over all $3+7+22+77+297$ partitions, from the definitions (Appendix A). (c) For $k \leq 6$ the values in (b) differ from $(k-1)(k-4) = -2, -2, 0, 4, 10$. They equal $T_{k-2} + 2 = 2, 3, 5, 8, 12$, and these exceed $(k-1)(k-4)$. For $k \geq 7$, $(k-1)(k-4) - (T_{k-2} + 2) = \frac{1}{2}((k-1)(k-6) - 4) > 0$. $\square$

10 The extremal partitions

For every $k \geq 5$ the explicit partition μ_k of Theorem 8.2 is extremal when $k \geq 7$. The extremal partitions for $k \leq 6$ are listed in Theorem 9.1(b). For $k \geq 7$ all of them are characterised as follows.

Theorem 10.1. Let $k \geq 7$, $D = (k-1)(k-4)$ and $\lambda \vdash T_{k-1} + 2$.

- (1) $d_{\mathcal{B}}(\lambda) = D$ if and only if $c_D \leq k-2$, i.e. $\mathcal{B}^{D-1}(\lambda)$ has at most $k-2$ parts.
- (2) If $k \geq 8$ and $d_{\mathcal{B}}(\lambda) = D$, then $\mathcal{B}^D(\lambda) = \zeta_{\{a,4\}}$ for some $a \in \{1, 3\}$. Moreover $(j(k-1), j(k-1) + j + 1, k-1)$ is a ramp for every $j = 1, \dots, k-4$, and conversely these ramps imply $d_{\mathcal{B}}(\lambda) = D$.

Proof. (1) If $c_D \leq k-2$, then $\lambda^{(D-1)}$ is not cyclic, since cyclic partitions have $k-1$ or k parts (eq. (3)). So $d_{\mathcal{B}}(\lambda) \geq D$, and equality holds by Theorem 7.1. Conversely let $d = d_{\mathcal{B}}(\lambda) = D$. In Case B of the proof of Theorem 7.1, $d \leq n-k+1 < D$, because $D - (n-k+1) = \frac{1}{2}k(k-7) + 1 > 0$. So Case A holds and $c_d = k-2$.

(2) Let $k \geq 8$ and $d = D$. Case B is excluded as in (1). Case A1 gives $d \leq n-k+2 < D$, because $\frac{1}{2}k(k-7) > 0$. Case A2 gives $d \leq \max\{k-1, k+3, n-k+1\} < D$. In the remaining case $b = \max E \geq 4$, (4) gives $D \leq (k-b)(k-1)$, so $b = 4$ and $E = \{a, 4\}$ with $a \leq 3$; $a \neq 2$ by Lemma 5.2(a). The ramp $(d, d+k-3, k-1)$ has length $L = k-3$ and $d = (L-1)(k-1)$. The equality part of Proposition 4.5 gives the ramps $(j(k-1), j(k-1) + j + 1, k-1)$, $j = 1, \dots, k-4$. Conversely, $j = k-4$ gives $c_D = k-2$, and (1) applies. $\square$

Remark 10.2 (computational observations, not used). For $k = 7$ the conclusions of (2) also hold, by an exhaustive check of the 1255 partitions of 23 with `bs.py`. The proof above does not cover $k = 7$, because Case A1 only gives $d \leq 18 = D$ there. For $7 \leq k \leq 11$, every extremal orbit in fact enters the cycle at $\zeta_{\{3,4\}}$. The numbers of extremal partitions are 66, 253, 1341, 6335, 28261 for $k = 7, \dots, 11$.

11 Where the straightforward extension of the C3 argument stops

The bound it yields. The Cell 3 argument (Griggs–Ho [4, Thm. 4.4]) applies verbatim to every non-triangular n of rank $k \geq 5$ and gives $D_{\mathcal{B}}(T_{k-1} + 2) \leq k^2 - 2k - 1$. This is strictly weaker than the truth for every $k \geq 7$:

$$k^2 - 2k - 1 - (k-1)(k-4) = 3k - 5 > 0.$$

Its losses are the same as described in Cell 4. The anchor t may lie several steps after d : the argument allows up to $k-2$, and by Lemma 5.2 the realised gap is $t-1-d = k-b-1 \leq k-3$. Also, ramps of value k cannot be excluded.

The next straightforward step (entry lemma plus ramp bound) is also not enough. Use Lemma 5.2 and (4), exactly as in Cell 4. Entries with $\max E \geq 4$ are handled. But there remain three entry points where no ramp gives the right bound:

- $E = \{1, 3\}$ and $E = \{2, 3\}$: the ramp at d has length $k-2$, so the ramp bound only gives $d \leq (k-3)(k-1)$. This exceeds $(k-1)(k-4)$ by $k-1$.
- $E = \{1, 2\}$: the last non-cyclic partition has $k-1$ parts, and $c_d = c_{d+1} = k-1$, so no ramp starts at d at all. For this entry only the Cell 3 bound $k^2 - 2k - 1$ is available.

So the straightforward extension stops at $k^2 - 2k - 1$, or at $(k-1)(k-3)$ if the $E = \{1, 2\}$ entries are ignored.

What our proof supplies instead. For these three entry points we do not use ramps at all. For $E = \{1, 2\}$ and $E = \{2, 3\}$ the entry partition μ has one or two very large parts ($k+1$, resp. k, k) and no other part within 2 of them. The pile-chain lemma (Lemma 6.1) then forces the largest pile to be N_1 or an original pile, i.e. a pile that has been shrinking by one per move since the start. For $E = \{1, 3\}$, μ also has a part $k-1$; there the sub-case analysis on c_{d-1} either reaches the same conclusion or produces a short ramp. Altogether $d \leq n - k + 2 = T_{k-2} + 3$ (the sub-case bound $k+3$ is at most $n - k + 2$ because $k \geq 5$). This bound is of order $k^2/2$ rather than k^2 , and it is at most $(k-1)(k-4)$ exactly when $k \geq 7$ (the identity $\frac{1}{2}k(k-7) \geq 0$; equality at $k = 7$). The mechanism is sharp for small k : for $2 \leq k \leq 6$ the partition 1^n , whose orbit enters at $\zeta_{\{1,2\}}$, has $d_{\mathcal{B}}(1^n) = n - k + 1 = T_{k-2} + 2$ and is extremal (exhaustive computation, Appendix A; for $k \geq 3$ this is also the Griggs–Ho witness [4, Thm. 4.5, case (2)]). This is why $k = 2, \dots, 6$ are exceptional.

12 Computations

Computations used in proofs: only Theorem 9.1(b) ($k = 2, \dots, 6$), exhaustive over the 406 partitions of 3, 5, 8, 12, 17 (Appendix A, well under a second). As sanity checks, `bs.py` (exhaustive for $n \leq 60$, about 1.5 minutes) confirms the following for $5 \leq k \leq 11$:

- Lemma 5.2, including the explicit μ in each case;
- the per-case bounds of Theorem 7.1, including the three sub-cases of Case A2;
- $D_{\mathcal{B}}(T_{k-1} + 2) = \max\{(k-1)(k-4), T_{k-2} + 2\}$;
- Theorem 10.1 (for $k \geq 7$).

It also confirms $d_{\mathcal{B}}(\mu_k) = (k-1)(k-4)$ for $5 \leq k < 80$. The maxima per entry type are sharp for $E = \{1, 2\}$ ($n - k + 1$) and for $\max E = b \geq 4$ ($((k-b)(k-1))$).

A Code for the exhaustive small cases

File `p3_small_cases.py` (Python 3, no dependencies; runs in well under one second). It computes $d_{\mathcal{B}}$ directly from the definition: the orbit of λ is followed until a partition repeats, and the index of the first occurrence of that partition is $d_{\mathcal{B}}(\lambda)$.

```
#!/usr/bin/env python3
"""Exhaustive small cases used in p3_c2/p3_c3/p3_c4/p3_c5.
Straight from the definitions: B(lambda), d_B(lambda) = number of steps before the
first partition that recurs in the orbit, D_B(n) = max over all partitions of n.
Runtime: well under one second."""

def partitions(n, maxp=None):
    maxp = n if maxp is None else maxp
    if n == 0:
        yield ()
        return
    for p in range(min(n, maxp), 0, -1):
        for rest in partitions(n - p, p):
            yield (p,) + rest

def B(lam):
    return tuple(sorted([len(lam)] + [x - 1 for x in lam if x > 1], reverse=True))

def d_B(lam):
    first = {}
    i = 0
    while lam not in first:
        first[lam] = i
        lam = B(lam)
        i += 1
    return first[lam] # the tail length: B^i(lam) is cyclic iff i >= this

def fmt(lam):
    # compact notation: (3,3,2,1) -> 3^2 2 1
    out = []
    for v in sorted(set(lam), reverse=True):
        m = lam.count(v)
        out.append(f"{v}^{m}" if m > 1 else f"{v}")
    return "(" + " ".join(out) + ")"

for n in [1, 3, 4, 5, 6, 7, 8, 9, 12, 17]:
```

```

parts = list(partitions(n))
d = {lam: d_B(lam) for lam in parts}
D = max(d.values())
ext = [lam for lam in parts if d[lam] == D]
print(f"n={n}: {len(parts)} partitions, D_B={D}, extremal: {'', '.join(fmt(l) for l in ext)}")

```

Output:

```

n=1: 1 partitions, D_B=0, extremal: (1)
n=3: 3 partitions, D_B=2, extremal: (1^3)
n=4: 5 partitions, D_B=2, extremal: (1^4)
n=5: 7 partitions, D_B=3, extremal: (1^5)
n=6: 11 partitions, D_B=6, extremal: (2^2 1^2)
n=7: 15 partitions, D_B=4, extremal: (1^7)
n=8: 22 partitions, D_B=5, extremal: (1^8)
n=9: 30 partitions, D_B=7, extremal: (3 2^2 1^2)
n=12: 77 partitions, D_B=8, extremal: (3^2 2^2 1^2), (1^12)
n=17: 297 partitions, D_B=12, extremal: (1^17)

```

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